Input power connnector

J1
ADAPTER 4.75–21V
PWR_FLAG
F1
FUSE TBD
VIN1
PGND

Battery connector

J3
BATTERY 2S/4S
BATTERY
PGND

Output power connnector

J2
VSYS OUTPUT
SYS
PGND

F1 rating/MPN OPEN; qualify source/cable hot-plug, polarity FET SOA and fuse coordination.

IC1
BQ25798RQMR

STAT_1	1	18	BATP
VBUS_1	2	19	BTST2
VBUS_2	3	20	PROG
BTST1	4	21	INT
REGN	5	22	BAT_1
D+	6	23	BAT_2
D−	7	24	SDRV_1
VAC2	8	25	SYS
VAC1_1	9	26	SW2
ACDRV2	10	27	GND
ACDRV1	11	28	SW1
QON	12	29	PMID
CE	13	30	STAT_2
SCL	14	31	VAC1_2
SDA	15	32	TS_2
TS_1	16	33	SDRV_2
ILIM_HIZ	17		

Q1 AON7528
Q2 AON7528
D1 12V
R1 270R
R4 100R
C23 47n 25V
R5 27k 1–2%
NT1
NT4
VIN1
VBUS
BATTERY
SYS
PMID
SDRV
STAT
BTST1
BTST2
REGN
CE_N
TS
ILIM_HIZ
GND
PGND

TP3 PGND
TP4 SYS
TP1 GND

boost side
BTST2
47n 25V C1
SW2
L1 2.2uH
buck side
SW1
47n 25V C2
BTST1

-INT
CHG_REQ
I2C_SCL
I2C_SDA
J4
MCU / BIAS INTERFACE
MAIN_3V3

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

REGN
R8 2k2
LED D2
STAT
C3 4.7u 10V
NT2
ILIM_HIZ
R6 5k6
R7 33k
TS
NTC temp sensor J6

C10 10u 25V
C4 10u 25V
C21 10u 25V
C19 10u 25V
C17 10u 25V
C14 10u 25V
C11 10u 25V
C5 100n 25V
C24 10u 50V
C25 10u 50V
C22 10u 50V
C20 10u 50V
C18 10u 50V
C15 10u 50V
C12 10u 50V
C6 100n 50V
C16 10u 50V
C13 10u 50V
C7 100n 50V
C8 1nF, 50V
C9 100n 50V
VIN1

REGN BAS116H D3 CE_BIAS
MAIN_3V3 BAS116H D4 CE_BIAS

CE_BIAS
R9 10k
CE_N
1R1
CHG_REQ
CHG_GATE
Q3 PMV16UN
R10 10k

Grounds

PGND PWR_FLAG
GND
GND PWR_FLAG
PGND = power ground (thicker traces, special net on PCB)
GND = digital ground (ordinary traces)
PWR_FLAGs are for DRC in PCB editor.

PMID bulk C20/C22/C24/C25: DNP in initial VSYS profiles; backup bulk is deferred.

PGND/GND remain separate; NT1 is the sole local tie beside IC1 pin 27. Keep switching loops short.

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Sheet: /
File: ups12v_v4.kicad_sch

Title: BQ25798 UPS module

Size: A3	Date:	**Rev: 2.1**
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